

AI-POWERED CREDIT RISK ASSESSMENT

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PROJECT GOAL

Develop a **machine learning model** that accurately assesses credit risk for loan applicants.

THE BUSINESS NECESSITY:

Effective credit risk assessment is essential to:

- **Minimize losses** from loan defaults
- **Make data-driven lending decisions**
- **Optimize portfolio performance** and capital allocation

This project moves us from traditional methods to predictive analytics.

PREDICTABLE FINDINGS: WHAT DRIVES DEFAULT RISK?



LOAN INTEREST RATE: THE AFFORDABILITY TRAP

High rates, often given to risky borrowers, create a vicious cycle.

They increase monthly burdens, making default more likely, not less. Our model identifies when a high rate itself becomes a risk factor.



HOUSING STATUS: A MEASURE OF STABILITY

OWN: Lowest risk. Indicates assets, stability, and strong cash flow.

MORTGAGE: Lower risk. Shows proven creditworthiness and commitment.

RENT: Higher risk. Can signal less disposable income and geographic transiency.



LOAN GRADE: THE VALIDATED SUMMARY

The model confirms and quantifies the traditional grade.

'A' Grade: Strong correlation with reliable repayment.

'G' Grade: High correlation with default, helping us price risk accurately.



SALARY/INCOME: THE REPAYMENT ENGINE

The #1 factor in repayment capacity.

The model focuses on the Debt-to-Income (DTI) ratio—the loan payment as a percentage of income. A high DTI is the strongest predictor of financial distress.

DEPLOYMENT-READY MODEL PERFORMANCE

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CHAMPION MODEL: RANDOM
FOREST CLASSIFIER

PERFORMANCE METRICS - ALL EXCELLENT

- Accuracy: 91.8% (9/10 predictions correct)
- Precision: 91.6% (Minimizes false alarms)
- Recall: 91.8% (Catches true defaulters)
- F1-Score: 91.7% (Perfect balance)

SUMMARY

From Data to Strategic Advantage

What We've Delivered:

- A high-performance risk assessment engine with **92% accuracy**
- **Production-ready** model files and documentation
- Clear **business use cases** and implementation plan

Strategic Impact:

- **Reduced financial losses** from defaults
- **Increased operational efficiency** in lending decisions
- **Competitive advantage** through superior risk pricing
- **Enhanced customer experience** with faster, fairer decisions





THANK YOU

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